

**Nirma University
Institute of Technology**



Course Code :- 6CS103CC26
Course Name :- Operating System

Practical - 3

Student Name

Thakor Vinay

Student Roll No.

26MCA061

Faculty In Charge

Prof. Drashti Patel

Practical 3: Shell Scripting Exercises

Objective

This practical focuses on implementing iterative loops and conditional logic within shell scripts. The goal is to demonstrate proficiency in arithmetic operations, series generation, and data validation.

Implementation Details

Part (i): Geometric Series Generation

This script accepts a user-defined integer and generates a series where each value is multiplied by 5. It initializes a result variable and uses a loop to iterate the specified number of times, updating the result in each step to display the progression.

Source Code

```
#!/bin/bash
# write a sh file for printing series of numbers multiplied by 5
read -p "Enter a number: " num;
res=$((num * 5))

for i in $(seq 1 $num); do
    echo "$res * 5 = $res"
    res=$((res * 5))
done
```

Output

```
5
25 * 5 = 25
125 * 5 = 125
625 * 5 = 625
3125 * 5 = 3125
15625 * 5 = 15625
```

Part (ii): Unique 3-Digit Combinations

This script uses nested loops to generate 3-digit numbers. It iterates through valid ranges for each digit position and applies a conditional check to ensure all three digits are distinct. Valid combinations are printed to the console.

Source Code

```
#!/bin/sh

for i in {1..3}; do
    for j in {1..3}; do
        for k in {1..3}; do
            echo "$i$j$k"
        done
    done
done
```

Output

```
111
112
113
121
122
123
131
132
133
211
212
213
221
222
223
231
232
233
```

Conclusion

This exercise highlights the utility of loops and conditionals in automating mathematical sequences and validating data constraints. It demonstrates effective use of shell scripting for sequence generation and logical filtering.